

5 Sentiment Analysis for given dataset • Negative Sentiment analysis • Positive Sentiment analysis

```
import pandas as pd
from textblob import TextBlob
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('location.csv')
df.head()
```

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	H
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	
2	2	2	Just finished an amazing workout! 🍌 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	
3	3	3	Excited about the upcoming weekend	Positive	2023-01-15	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	

Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

```
df['polarity'] = df['Text'].apply(lambda x: TextBlob(x).sentiment.polarity)
df.head()
```

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	H
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	
2	2	2	Just finished an amazing workout! 🍌 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	
3	3	3	Excited about the upcoming weekend getaway! ...	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	
4	4	4	Trying out a new recipe for dinner tonight. ...	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Cooking #Food	12.0	25.0	Australia	2023	1	15	

Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

```
ans = []
for i in df['polarity']:
    if i > 0.5:
        ans.append('positive')
    else:
        ans.append('negative')

df['pred'] = ans
```

```
df.head()
```



	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	H
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	
2	2	2	Just finished an amazing workout! 🍌 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	
3	3	3	Excited about the upcoming weekend getaway! ...	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	
4	4	4	Trying out a new recipe for dinner tonight. ...	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Cooking #Food	12.0	25.0	Australia	2023	1	15	

Next steps:

[Generate code with df](#)

[View recommended plots](#)

[New interactive sheet](#)

```
pos_tweets = df[df['pred'] == 'positive']
neg_tweets = df[df['pred'] == 'negative']
```

```
sns.kdeplot(pos_tweets['polarity'], label='Positive', shade=True)
sns.kdeplot(neg_tweets['polarity'], label='negative', shade=True)
plt.xlabel("Polarity Score")
plt.ylabel("Density")
plt.title("Sentiment Analysis of Tweets")
plt.show()
```

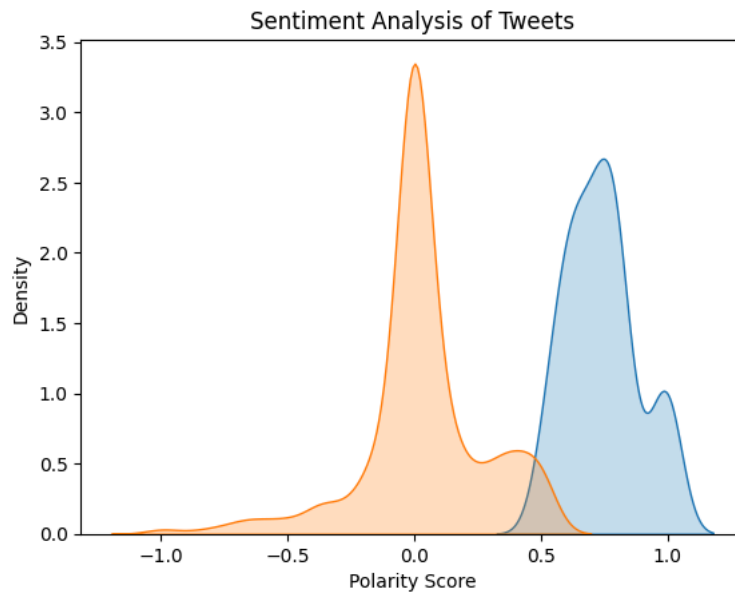
```

<ipython-input-16-07139124ef25>:1: FutureWarning:
`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

sns.kdeplot(pos_tweets['polarity'], label='Positive', shade=True)
<ipython-input-16-07139124ef25>:2: FutureWarning:
`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

sns.kdeplot(neg_tweets['polarity'], label='negative', shade=True)

```



```

ans = []

for i in df["polarity"]:
    if i>0.1:
        ans.append("positive")
    elif(i<-0.1):
        ans.append("negative")
    else:
        ans.append("neutral")

df["sentiment_new"] = ans

```

```

sentiment_counts = df['sentiment_new'].value_counts()
print(sentiment_counts)

```

```

<ipython-input-16-07139124ef25>:3: FutureWarning:
sentiment_new
neutral      394
positive     240
negative      98
Name: count, dtype: int64

```

```

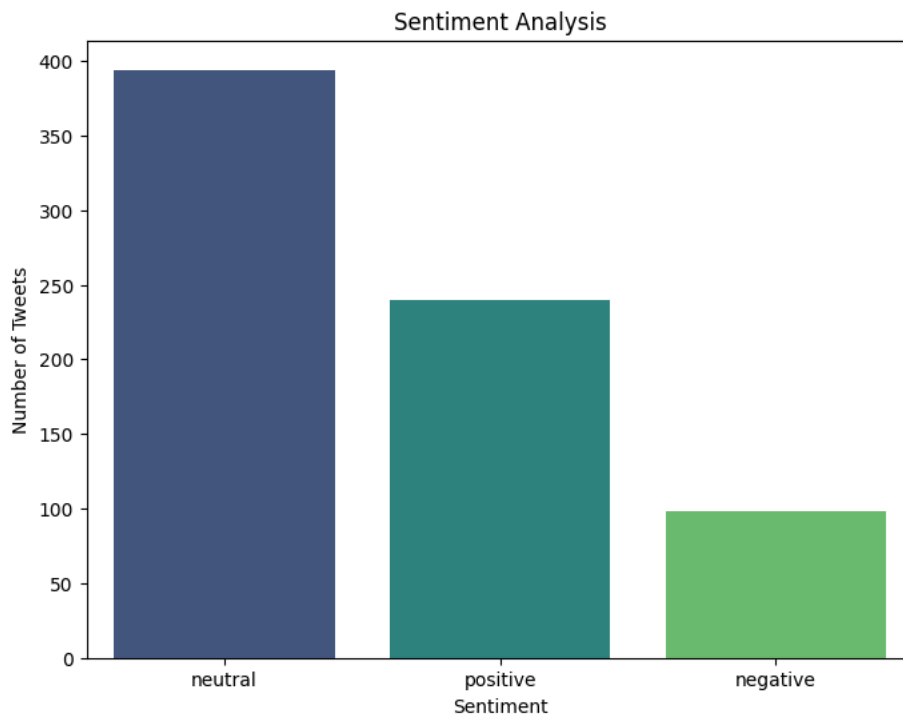
plt.figure(figsize=(8, 6))
sns.barplot(x=sentiment_counts.index, y=sentiment_counts.values, palette="viridis")
plt.xlabel('Sentiment')
plt.ylabel('Number of Tweets')
plt.title('Sentiment Analysis')
plt.show()

```

```
<ipython-input-20-80a1c75b7baa>:2: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `le

```
sns.barplot(x=sentiment_counts.index, y=sentiment_counts.values, palette="viridis")
```



```
df["pred"]
```



```

      pred
0  positive
1  negative
2  positive
3  negative
4  negative
...      ...
727 positive
728 positive
729 positive
730 positive
731 positive

```

732 rows × 1 columns

df.index object

```

pos_tweets = df[df["sentiment_new"] == "positive"]
neg_tweets = df[df["sentiment_new"] == "negative"]
neutral_tweets = df[df["sentiment_new"] == "neutral"]

```

```

print("Positive Tweets:")
for tweet in pos_tweets['Text'].head():
    print("-", tweet)

print("\nNegative Tweets:")
for tweet in neg_tweets['Text'].head():
    print("-", tweet)

print("\nNeutral Tweets:")
for tweet in neutral_tweets['Text'].head():
    print("-", tweet)

```



```

Positive Tweets:
- Enjoying a beautiful day at the park!

```

- Just finished an amazing workout! 🏋️
- Excited about the upcoming weekend getaway!
- Trying out a new recipe for dinner tonight.
- The new movie release is a must-watch!

Negative Tweets:

- Traffic was terrible this morning.
- Feeling grateful for the little things in life.
- Missing summer vibes and beach days.
- Exploring the city's hidden gems.
- Reflecting on the past and looking ahead.

Neutral Tweets:

- Rainy days call for cozy blankets and hot cocoa.
- Political discussions heating up on the timeline.
- Feeling a bit under the weather today.
- Late-night gaming session with friends.
- Attending a virtual conference on AI.

```
sns.kdeplot(pos_tweets["polarity"], shade=True, label="Positive")
sns.kdeplot(neg_tweets["polarity"], shade=True, label="Negative")
sns.kdeplot(neutral_tweets["polarity"], shade=True, label="Neutral")
plt.xlabel("Polarity Score")
plt.ylabel("Density")
plt.title("Sentiment Analysis of Tweets")
plt.legend()
plt.show()
```

 <ipython-input-24-ff68fa26d5ad>:1: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

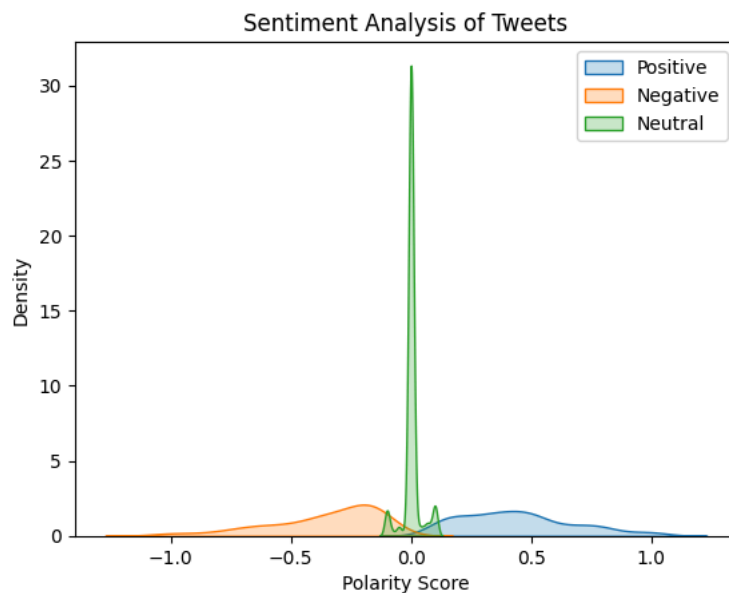
```
sns.kdeplot(pos_tweets["polarity"], shade=True, label="Positive")
<ipython-input-24-ff68fa26d5ad>:2: FutureWarning:
```

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

```
sns.kdeplot(neg_tweets["polarity"], shade=True, label="Negative")
<ipython-input-24-ff68fa26d5ad>:3: FutureWarning:
```

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

```
sns.kdeplot(neutral_tweets["polarity"], shade=True, label="Neutral")
```



Start coding or [generate](#) with AI.

Start coding or [generate](#) with AI.